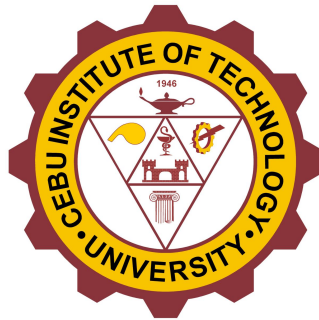


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CSIT 238 - HUMAN-COMPUTER INTERACTION

Digital Identity and Accessibility: Addressing Challenges of Persons with Disabilities

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ABSTRACT

Discrimination in obtaining a digital identity is one of the accessibility challenges that people with disabilities (PWDs) must deal with. Authentication systems like CAPTCHAs, liveness checks, biometrics, and multi-factor verification often assume typical ranges of vision, hearing, speech, and physical or cognitive ability. Recent statistics show that there are currently about 1.57 million registered PWDs in the Philippines. The goal of this study is to capture Filipino PWDs' experiences when engaging with digital identity mandates and infrastructures including transactions with online government services. Twenty participants each for the survey and five for the interviews were sourced from Region 7 in March of 2026. Qualitative analysis was thematically conducted on interview transcripts, and quantitative descriptive statistics were used on survey data. Results indicate that PWDs in the Philippines face authentication systems that are often unable to account for their user experiences and desired an authentication feature that would adequately balance security with accessibility. Research into and design inspiration for an accessible digital identity feature was included to help close the gap towards a multimodal solution.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Digital identity is the key to unlocking essential services like healthcare, social welfare, voting, banking, and emergency aid. When identity systems are inaccessible, people with disabilities are unfairly excluded from these vital services. Existing systems were largely designed without input from disabled users, resulting in interfaces that assume standard visual, auditory, and motor capabilities. According to Cropley et al (2026), individuals with disabilities face significant barriers during authentication processes, often finding them difficult, frustrating, or nearly impossible. Using Constructivist Grounded Theory to analyze interviews (n = 15), the findings reveal that accessibility barriers hinder effective authentication, creating a societal gap in ease-of-use compared to non-disabled users. Participants emphasized the importance of balancing usability and security, indicating that future accessible authentication systems must ensure both inclusivity and protection of user information. A study indicates that to close accessibility gaps and guarantee fair participation in global digital economies, digital identification systems need to change (Alhassan, 2024) ([SSRN link](#)). When considered as a whole, these results highlight the fact that inaccessible identification systems are a global problem that calls for inclusive design principles.

There are an estimated 1.57 million registered persons with disabilities (PWDs) in the Philippines, yet they still have limited access to digital government services and identity systems. PWDs encounter major obstacles when attempting to prove their identity online as public services, financial institutions, and social welfare programs move more to digital platforms. People with visual, hearing, motor, and cognitive

impairments are systematically excluded from government offices due to their physical inaccessibility, lack of assistive technology integration, and complicated multi-step identification procedures. People with disabilities are unable to completely utilize digital services due to accessibility issues on Philippine e-government websites (Bokingito et al., 2025). To prevent fraud and improve verification, the Department of Social Welfare and Development and the National Council on Disability Affairs are now using a PWD ID system. This system has both physical versions and uses a QR code for authentication (DSWD & NCDA, 2025). However, people with disabilities still face problems when trying to use services. They are often stopped from participating because usual security methods like CAPTCHAs, multi-step verification, and biometric scans are not easily available. This emphasizes the conflict between security and usability since PWDs want both protection of their personal information and convenience of use, but existing systems frequently compromise one for the other (World Wide Web Consortium, 2023). Physical inaccessibility of government offices, lack of assistive technology integration, and complex multi-step authentication processes create systematic exclusion for individuals with visual, hearing, motor, and cognitive impairments. Addressing this gap is both a human rights obligation under the Magna Carta for Disabled Persons (RA 7277) and a practical necessity for inclusive national digital transformation.

This study aims to investigate the challenges faced by persons with disabilities (PWDs) in accessing digital identity systems in the Philippines, to design a more inclusive, accessible, and multi-modal digital identity solution that accommodates diverse physical, sensory, and cognitive needs.

1.2 Background of the Study

As access points to banking, healthcare, education, and government services, electronic identity (e-ID) systems have become essential to modern life. With the goal of guaranteeing that users can access digital platforms safely, these systems are made to offer secure authentication and identity verification. E-ID programs like the Unified PWD ID and the Philippine Identification System (PhilSys) are a part of larger initiatives in the Philippines to improve service delivery.

Current e-ID systems rely primarily on traditional authentication methods including passwords, CAPTCHAs, biometrics, and multi-step verification. People with disabilities (PWDs) are at a disadvantage since these methods often assume standard visual, auditory, and motor abilities, even though they function well for the broader public. Many PWDs suffer from frustration, reliance on caretakers, or total exclusion from digital services as a result. This emphasizes the urgent need for accessible authentication systems that accommodate a variety of physical and cognitive abilities while finding a balance between usability and security.

PWDs have significant challenges when utilizing these systems. Image-based CAPTCHAs and poorly coded interfaces that screen readers cannot decipher are difficult for visually challenged users. Systems that rely on audio verification exclude people with hearing impairments. Complex passwords and biometric gestures are challenging for users with motor disabilities, while multi-step authentication processes are overwhelming for individuals with cognitive impairments. Infrastructure issues like inadequate internet access and a lack of assistive device integration exacerbate exclusion even more than interface design.

There is limited research on workable, established remedies specific to the Philippine setting, despite the fact that both local and international studies draw attention to these accessibility issues. The work currently in publication frequently notes obstacles but does not offer or assess inclusive prototypes. Design frameworks and empirical research that include accessibility elements into authentication systems while striking a balance between security and usability are required. By closing this gap, the difficulty gap between impaired and non-disabled users would be reduced and PWDs would be able to independently and fairly access necessary digital services.

1.3 Problem Statement

This set of questions contains a framework for exploring the lived experiences of persons with disabilities (PWDs) in the Philippines when using digital identity systems.

1. What are the current experiences and challenges of PWDs when attempting to access or use digital identity systems in the Philippines?
2. What specific interface, technological, or procedural barriers most frequently prevent PWDs from independently completing digital identity verification?
3. What assistive tools, workarounds, or caregiver-mediated processes do PWDs currently rely on when official identity systems are inaccessible?
4. How do environmental and infrastructural factors (e.g., lack of internet, absence of assistive device support, physical location) compound digital identity access challenges for PWDs?
5. How do PWDs and their caregivers perceive the severity, impact, and urgency of digital identity exclusion on their ability to access essential public services?

1.4 Objectives of the Study

With the aim to find design opportunities for a more inclusive, accessible, and multimodal digital identity solution, it is necessary to investigate the obstacles and difficulties that people with disabilities encounter while trying to utilize digital identity systems in the Philippines.

Specific Objectives:

1. To identify the most critical technological and design barriers that prevent PWDs from independently accessing digital identity systems.
2. To examine how existing identity platforms fail to accommodate diverse disability types, including visual, auditory, motor, and cognitive impairments.
3. To determine the workarounds, coping strategies, and assistive tools currently used by PWDs and their caregivers when official identity systems are inaccessible

1.5 Significance of the Study

The findings of this study are beneficial to the following individuals. It can help improve accessibility, usability, and inclusiveness in digital identity systems by addressing the barriers faced by people with disabilities (PWDs).

- **For users (PWDs):** Digital identification systems that are accessible promote self-sufficiency, lessen dependency on caretakers, and guarantee equitable access to vital services like banking, healthcare, and welfare. By reducing obstacles that now prevent or irritate impaired people, they directly increase the quality of life.
- **For designers:** Studying these challenges provides actionable insights into inclusive design principles. Designers gain a clearer understanding of how to build authentication workflows that accommodate diverse abilities, leading to more user-friendly, secure, and equitable systems.

- **For the HCI community:** This research highlights how human-computer interaction (HCI) intersects with social inclusion, offering both theoretical and practical approaches to make digital systems equitable. It shows that accessibility must be treated as a central design priority, not an afterthought, to ensure fairness in digital societies.

CHAPTER 2: Literature review of related literature.

The chapter provides a thematic literature review as pertains to the related literature on access to digital identity, authentication, and inclusive design that accommodates persons with disabilities (PWDs). The review is divided into three themes which are directly pertinent to the study: (1) Accessibility and Inclusive Design, (2) Interaction Techniques to a Diverse User, and (3) Usability and User Experience with PWDs. Ten academic sources are considered, 7 of which were published in 2021-26.

2.1 Thematic Review

Theme 1: Access and Design.

Cropley, Whittington, and Dogan (2026) explore issues that people with disabilities face when using electronic identity authentication systems. They find that the existing authentication and e-ID systems are often challenging, distressing and inapplicable to disabled users posing a great challenge at making them independent digital participants. They suggest establishing safe, non-discriminatory, and easy-to-use authentication methods, especially, the possible potential of biometrics and new passwordless systems as attractive options to make access to digital identity more fair.

According to Bokingito, Beloy, Ecleo, Alce, Borinaga, and Galido (2025), the authors investigate the availability of Philippine e-government websites and their usefulness in providing services to people with disabilities. According to their study, the government websites have a certain level of accessibility, but they do not possess complete functionality, integration of digital services and ease of use by disabled members of the society. They suggest drastic development in digital services, mobile access, and system integration to have a fully inclusive e-government experience that

ensures that all Filipino citizens, including persons with disabilities, are equally served.

Putting a particular focus on digital accessibility, Nacheva and da Costa (2024) promptly state that, in the context of inclusive education, the learning resources should be deliberately created in a way that enables all students to engage equally irrespective of their capabilities. Their analysis demonstrates that online learning materials are often not up to the full packs of accessibility, thus posing a barrier to disabled learners. They suggest that it is essential to comply with known accessibility guidelines, regularly test online content, and implement an inclusive systems design to ensure that each user has an equal chance to learn.

Furnell, Helkala, and Woods (2022) emphasize authenticating to be a major barrier to access by users with disabilities. They find that existing authentication systems are user-friendly to the common person, and people with various physical, sensory, and cognitive disabilities are excluded. They suggest that authentication procedures should be adaptable, easy and accommodative to make sure that every user can access digital services safely and autonomously.

Furnell, Helkala, and Woods (2021) discuss the systematic disfavoring of individuals with disabilities in the contemporary cybersecurity systems established without considering diverse users. They find that the existing authentication systems cater to the needs of the average user, and disablement leaves disabled people at risk of exclusion and insecurity. They suggest that authentication systems be re-architected in a way supportive of inclusivity as a matter of default so that security can be provided to all users irrespective of their physical, sensory, and cognitive capabilities.

Theme 2: Interaction Methods with different users.

Mudiyansele and Weeraskara (2025) examine the barriers to the accessibility of the current electronic identification systems and the effects that they have on individuals with disabilities. Their research indicates that the current e-ID systems are not entirely accessible and pose a serious obstacle that makes disabled people unable to enjoy the same benefits of digital services. They suggest the creation and experimentation of more inclusive e-ID system that eliminates these obstacles and guarantees equal opportunities to all users in terms of their digital accessibility.

Di Campi and Luccio (2025) suggest that the existing authentication systems do not address the requirements of users with cognitive disabilities, as they tend to be overly complex and, therefore, fail to provide good usability and security. Their research mentions biometric means as the most promising solution to provide more access and inclusion towards this population in terms of authentication. They suggest moving to more accommodating and less demanding authentication methods especially biometrics to suit the different cognitive capacities that people with disabilities have.

A systematic literature review by Cropley, Whittington, and Dogan (2024) investigated how authentication systems could be improved to better meet the needs of users with disabilities. Their results indicate that existing authentication systems always fail disabled individuals because they are more focused on mainstream functionality, rather than inclusive design. They suggest creating effective, user-friendly authentication systems that would balance between security and convenience, as well as the significance of high motivation in an organization to lead to inclusive innovation.

Theme 3: PWDs Usability and User Experience.

Furnell, Helkala, and Woods (2022) also emphasize that authentication is another major usability barrier to people with disabilities, especially when the systems are designed with a set of various visual, auditory, and motor abilities. Their study has shown that the inaccessible systems not only frustrate users but also compromise their freedom of accessing digital services independently and safely. They suggest that usability testing should have participants of varying disabilities in order to make sure that authentication solutions are indeed fair.

Helkala (2012) shows that conventional authentication procedures are very inappropriate to people with disabilities because they do not support a wide range of different physical and cognitive abilities. The paper highlights that authentication systems need to be re-engineered in a more user-friendly way to accommodate individuals with certain disabilities, allowing them to access online services safely and autonomously. Helkala suggests that more convenient and adaptable authentication systems should be developed, which will be adapted to the different capabilities of people with disabilities.

2.2 Research Gap

Although the current literature has managed to clearly show the barriers to accessibility in the authentication and digital identity systems, there are still considerable gaps in the literature. The majority of studies are conducted with general populations in the Western context and there are very few studies that concentrate on particular issues that persons with disabilities (PWDs) encounter in the Philippine context. Moreover, the available literature records more issues than offers or empirically investigates any solutions, inclusive in nature, which can effectively solve

the more than two types of disability, such as visual, auditory, motor and cognitive deficits. There are also minimal studies on how environmental and infrastructural aspects such as low internet connectivity and integration of assistive devices exacerbate digital identity exclusion in developing nations. This void underscores the necessity of a study that does not only examine the lived experiences of Filipino PWDs in using digital identity systems, but also suggests and analyzes a multimodal, inclusive authentication system that addresses the various needs of the PWDs. The current research fills this gap by integrating qualitative exploration of user experiences with the design of an accessible digital identity feature that promotes both security and usability to all users.

CHAPTER 3: METHODOLOGY

3.1 Research Design

For this study, the researchers used a mixed-methods approach. This was chosen because the topic required both numbers and stories — survey data to see how widespread the barriers were, and interviews to actually understand what those barriers felt like for the people experiencing them. A purely quantitative study would have missed too much context, and an entirely qualitative one would not have been enough to show patterns across a larger group. Combining both allowed the researchers to cross-check findings and feel more confident that what they found was accurate.

According to Creswell and Plano Clark (2018), mixed methods works well when neither quantitative nor qualitative data alone is sufficient to answer the research questions. That is exactly the situation here. The survey gave the team a broad picture, while the interviews filled in the gaps. When findings from both matched up, it gave the researchers more reason to trust the results.

3.2 Participants

The researchers gathered data from two groups. The first was a survey group made up of 20 respondents, all from Region 7 (Central Visayas) in the Philippines. To be included, participants had to have a disability — whether formally diagnosed or self-reported — and had to have used a digital identity or government service platform at least once before. Ages ranged from 18 to 37, and the group was split evenly between male and female respondents (50% each). Most of them, 60% or 12

out of 20, identified as having a visual disability. The rest had hearing disabilities (20%), preferred not to say (15%), or had a motor disability (5%). One notable finding even at this stage: all 20 said they do not rely on a caregiver when using digital services, meaning they navigate these systems on their own.

The second group was five individuals who took part in longer one-on-one interviews. The researchers chose them on purpose — what Patton (2002) calls purposive sampling — to make sure different types of disabilities were represented. There was one participant with ADHD (a first-year student), one with low vision ranging from 700 to 800+ diopters (a nursing student), one with a visual impairment, and two with hearing-related difficulties. All five were regular users of digital identity platforms, and all agreed to share their personal experiences openly.

3.3 Research Instruments

The team used two instruments to collect data: an online survey and an interview guide.

The survey had 10 questions in total. The first few items asked for basic demographic information — things like age, sex, region, and disability type. From there, the questions shifted toward the participants' actual experience: how often they ran into problems using digital identity systems, what specific barriers they encountered, how severe those barriers felt on a scale of 1 to 5, whether they needed someone else's help to get through the process, and how satisfied they were overall. It was all done online so participants could complete it on their own without needing assistance.

For the interviews, the team prepared a semi-structured guide — meaning there were set questions, but the conversation was allowed to go wherever the participant took it. The questions asked about their day-to-day experiences with digital identity systems, what gets in the way, how they usually deal with those problems, and what they wish designers would do differently. The researchers made a conscious effort not to just check off questions but to actually listen, especially for emotional cues and the kinds of workarounds people quietly rely on. All five interviews were done through video call.

3.4 Procedure

Data collection happened in two phases, both done in March 2026. The first phase was the survey. The researchers reached out to potential participants in Region 7 who had disabilities and had used digital identity or government service platforms before. Once they agreed to join, the survey link was sent to them digitally. Out of everyone who responded, 20 valid answers were collected. Each response was checked to make sure it was complete before the data was brought together for analysis.

The second phase was the interviews. Five participants were selected — some from those who took the survey, others through additional contacts — making sure to include different disability types. Each interview was set up individually through video call. The researchers followed the guide but stayed flexible, letting the conversation go deeper when something important came up. Sessions were documented so the team could go back and review them during analysis.

3.5 Data Analysis Plan

For the survey, the researchers looked at the numbers through descriptive statistics. This meant calculating how many respondents fell into each category — disability type, sex, region, what barriers they faced, whether they needed a caregiver — and turning those into percentages. For the Likert-scale questions on severity and satisfaction, the team calculated mean scores. Both came out at 2.9 out of 5. Charts and graphs were used to present these results more clearly.

The interview data was handled through thematic analysis, following the approach laid out by Braun and Clarke (2006). The team went through all the interview notes and responses and started looking for patterns — things that kept coming up, even when participants had different disabilities. Themes around barriers, coping strategies, emotional reactions, and design suggestions emerged pretty consistently. What made the researchers more confident in these findings was that they matched up well with the survey data. For example, 60% of survey respondents had visual disabilities, and three of the five interviewees fell into that same category. The same core issues — unclear audio or visual instructions, and processes that felt too complicated — showed up in both sets of data. Actual quotes from the participants were kept to bring the findings to life and make sure the analysis stayed grounded in real experiences.

Declaration on Generative AI

The use of generative AI was limited to language refinement and did not replace critical thinking, original writing, or scholarly judgment. The researchers take full responsibility for the authenticity, validity, and integrity of this work.

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CONTRIBUTION MATRIX

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